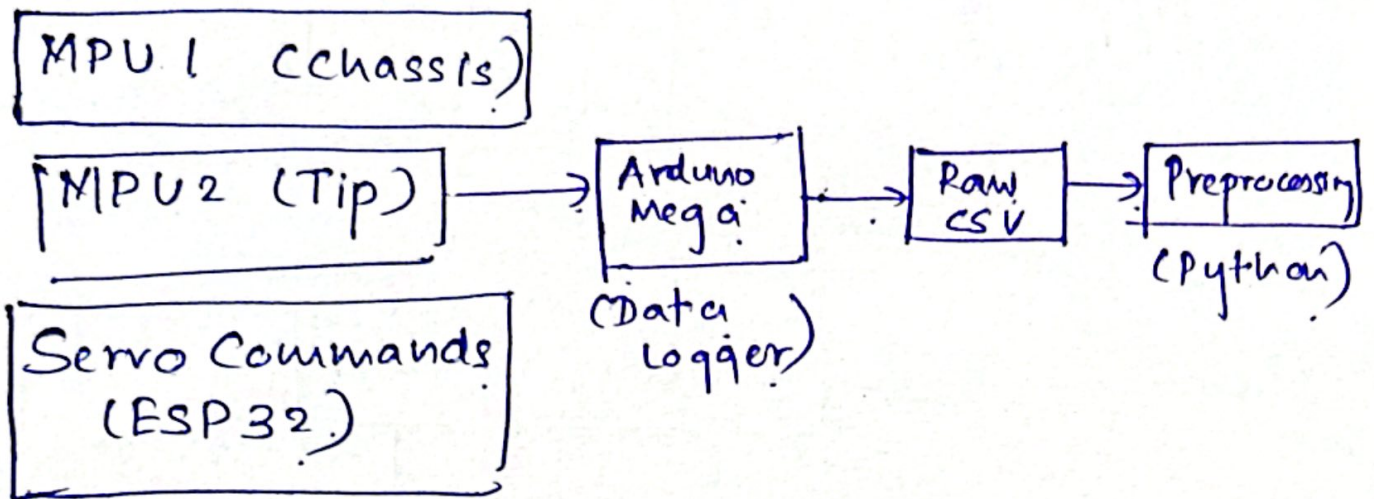
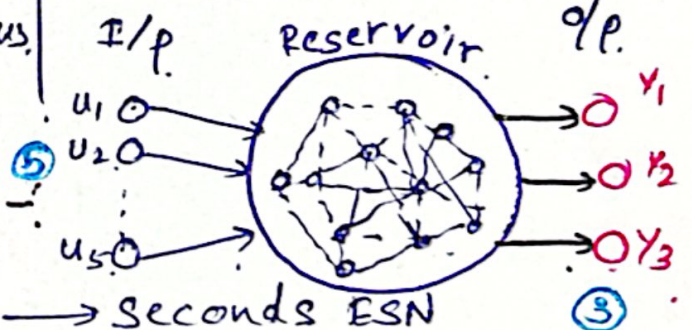


Model Training (ESN)



Raw Data Format - CSV File.
(For offline testing)

time.(ms)	mpu1-ax	mpu1-ay	mpu1-az	mpu2-ax	mpu2-ay
mpu2-az	mpu2-ax Servo1 (ms)	Servo2-us	I/p	Reservoir	O/p



Steps for preprocessing -

- * Time conversion, ms $\rightarrow$ seconds ESN
- * Scale MPU 6050 data (counts $\rightarrow$ g)
- * Map MPU1 $\rightarrow$ Chassis
MPU2 $\rightarrow$ Tip
- * Select inputs and outputs:
Inputs $\rightarrow$ Servo1, Servo2, Chassis Ax, Chassis Ay, Chassis Az

Outputs $\rightarrow$ Tip Ax, Tip Ay, Tip Az

- * Normalize $\rightarrow$ Inputs
 $\rightarrow$ Outputs

* Split datasets.

- Train. (no touch)
- Validation. (without touch)
- Test.
 - With touch
 - Without touch

Notes

- Washout periods (Ignored initial samples until reservoir warms up) (200 samples).
- Leak rate - Control of how much past inter. remains. (0.3)
- Reservoir size (N) (Testing) 500.
- Activation - tanh.
(Limits Neutone outputs to -1 to +1 while adding non-linearity)

Residual and Touch Detection

(1) Residual (per axis)

$$e(t) = y_{\text{actual}}^{(t)} - y_{\text{predicted}}^{(t)}$$

(4) Threshold

$$\text{touch}(t) = \begin{cases} 1; & r_{\text{filt}}^{(t)} > Th. \\ 0; & \text{otherwise.} \end{cases}$$

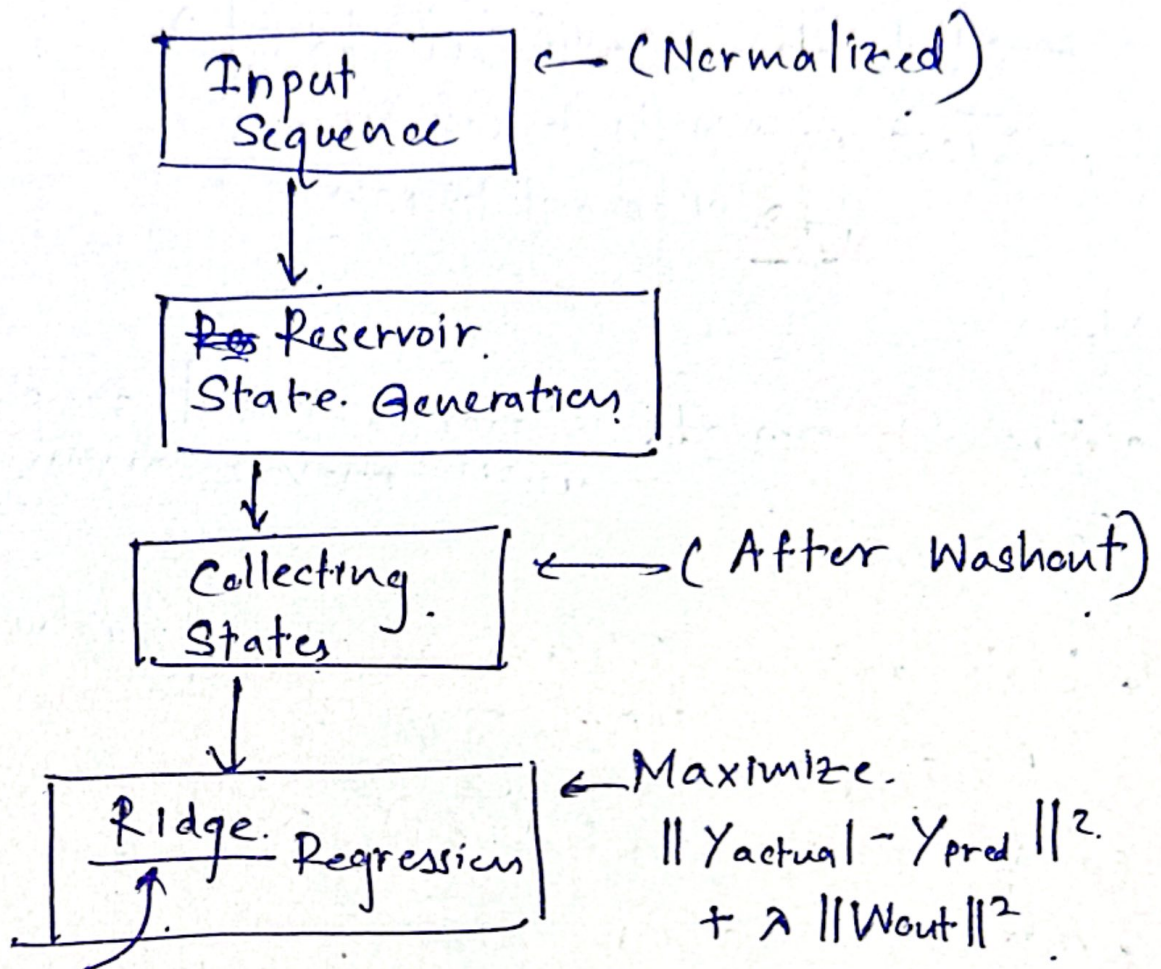
(2) Residual magnitude.

$$r(t) = \sqrt{e_x^2 + e_y^2 + e_z^2}$$

(3) Half wave rectification.

$$r_{\text{pos}}(t) = \max(r(t), 0)$$

Training process.



(Output weight calculation)

- Fits the reservoir states to the target sensor data. (Prevents large weights, improve prediction accuracy) and reduce overfitting)

Parameters

- Reservoir size (N) = 500
- Leak Rate (α) = 0.3
- Spectral radius (ρ) = 0.9
- Input scaling (β) = 0.1
- Ridge parameter (λ) = $1e-6$